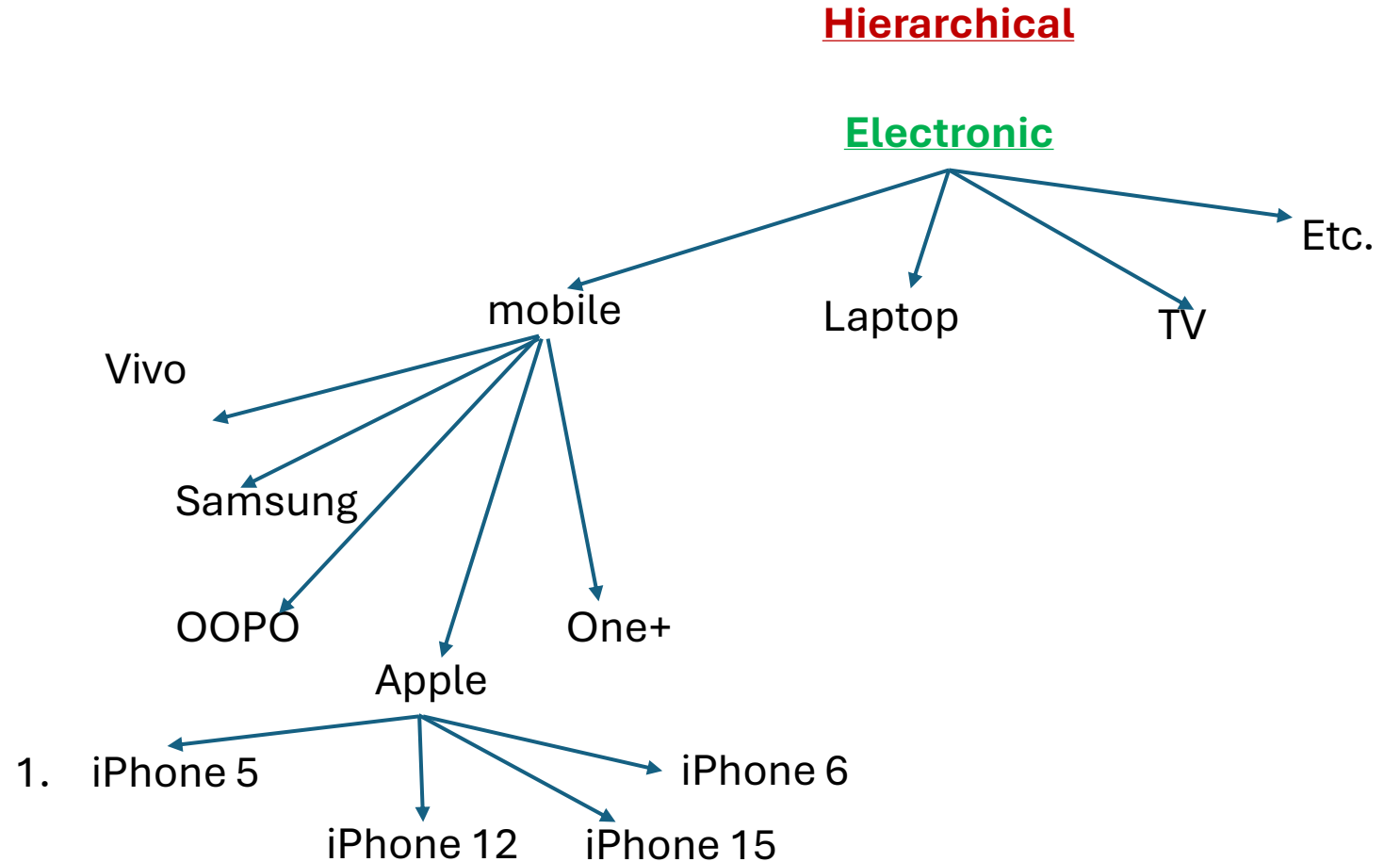
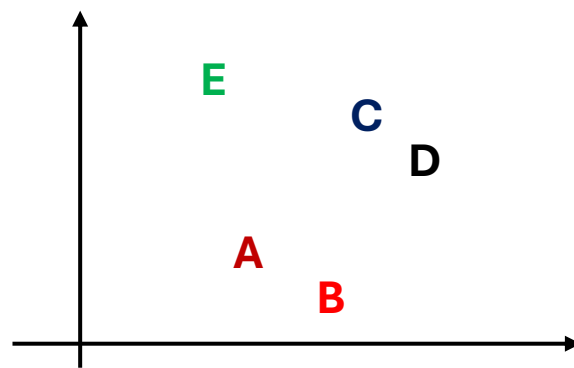
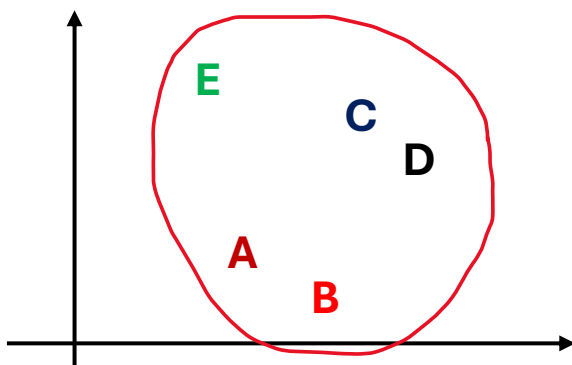
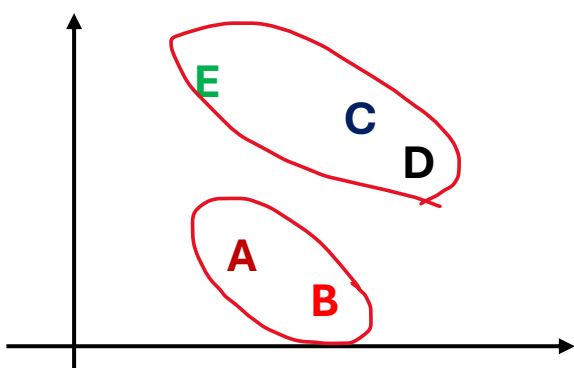
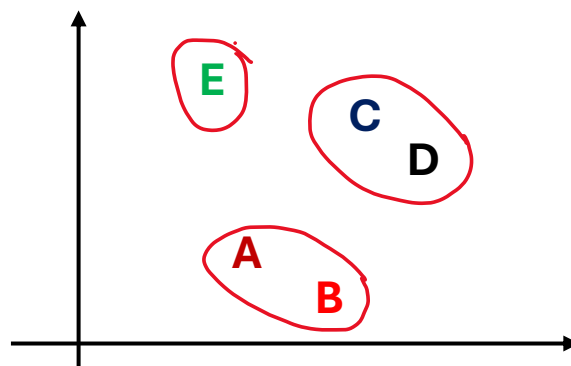
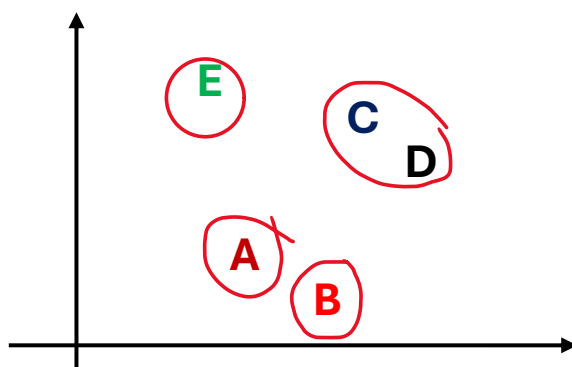
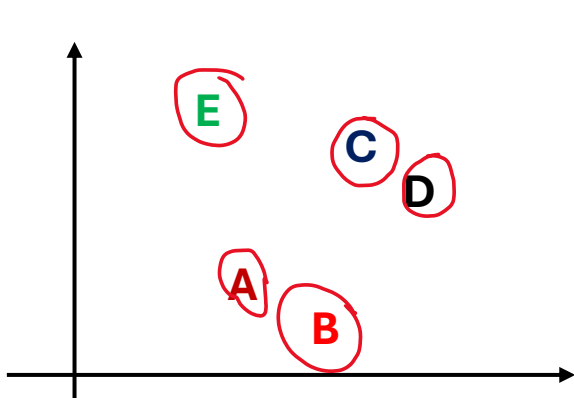
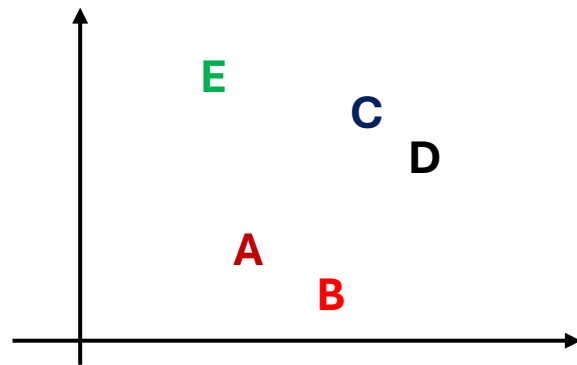
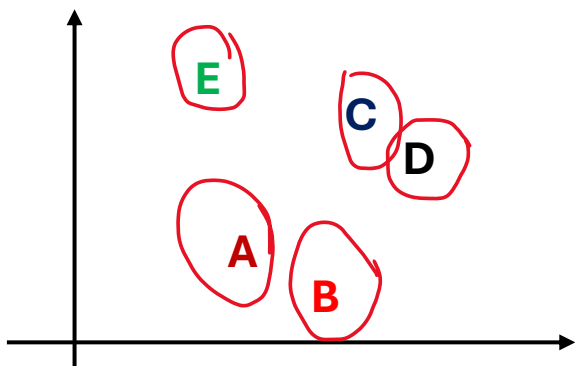
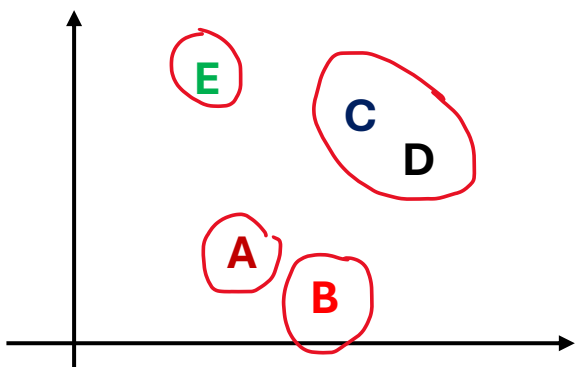
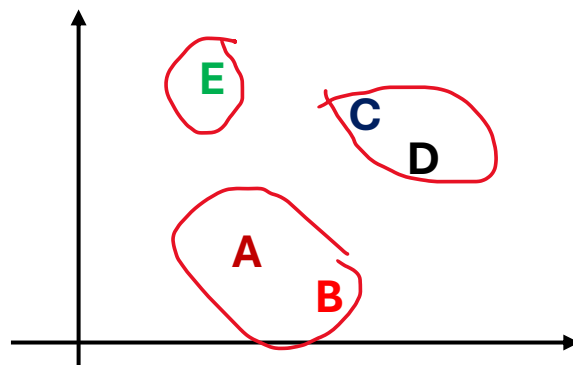
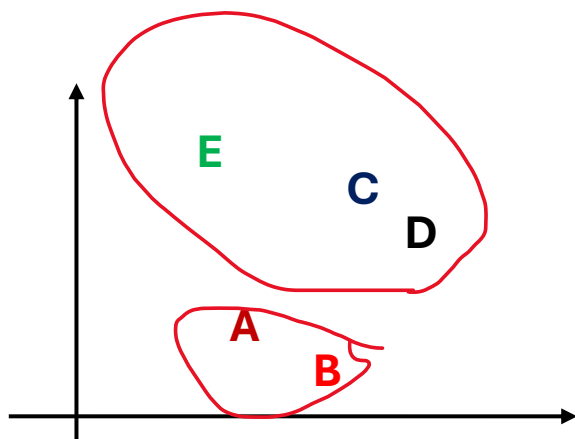
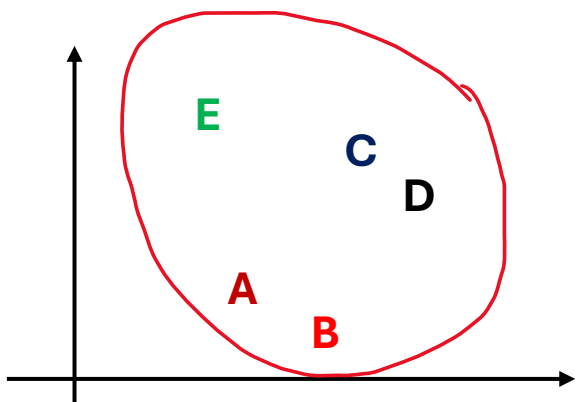


Hierarchical Clustering

1. Agglomerative
2. Divisive

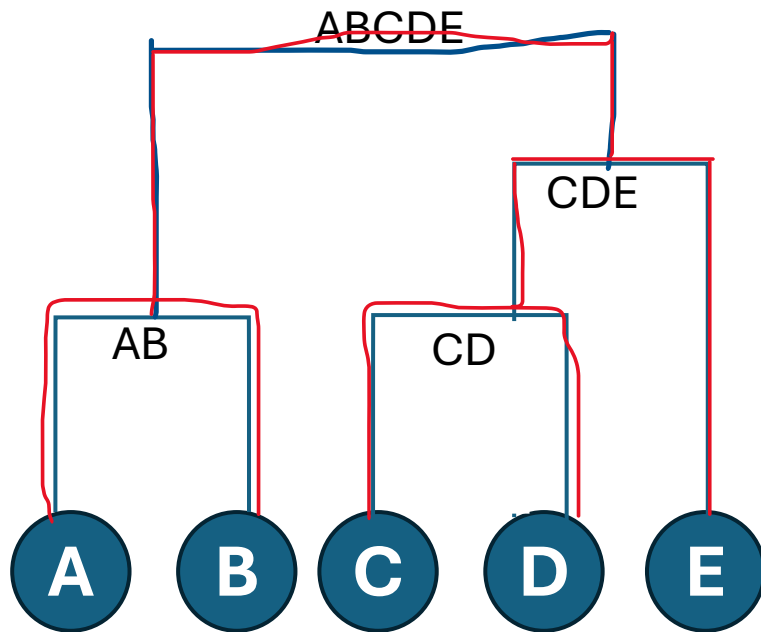
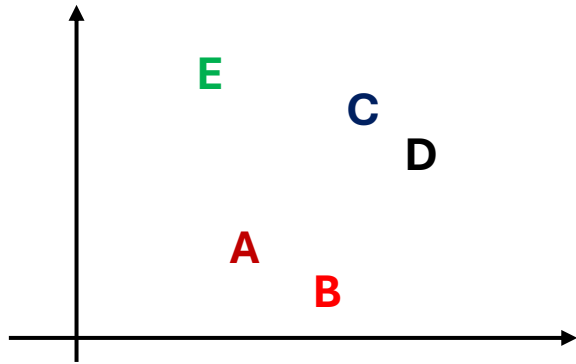






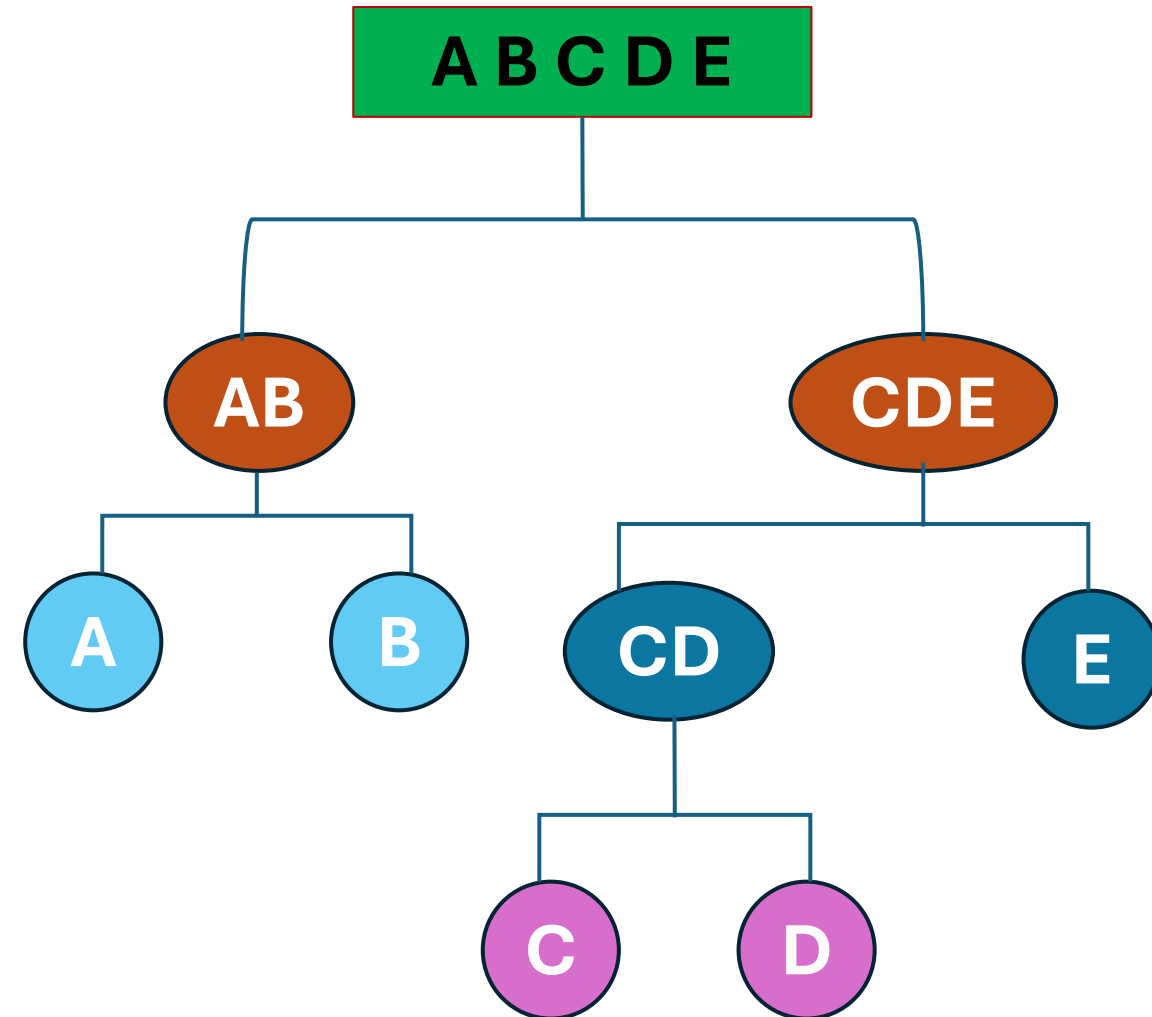
1. Agglomerative

B C D E



Bottom to UP

2. Divisive



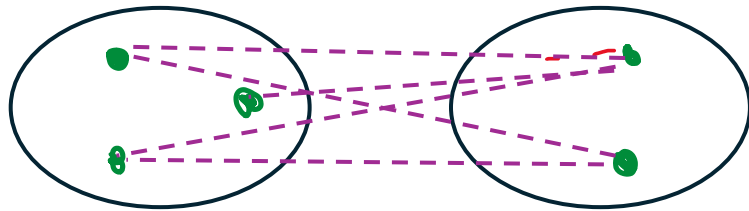
UP TO Bottom

DENDROGRAM

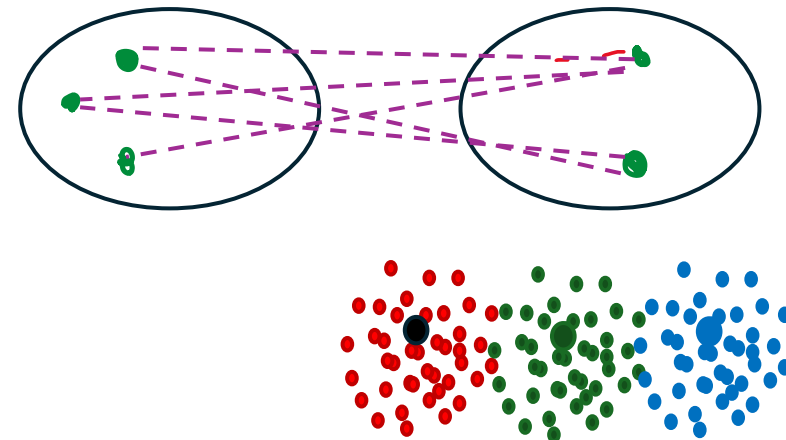
Types Of Agglomerative Clustering

1. Min (Single –link)
2. Max (complete Link)
3. Average
4. Ward

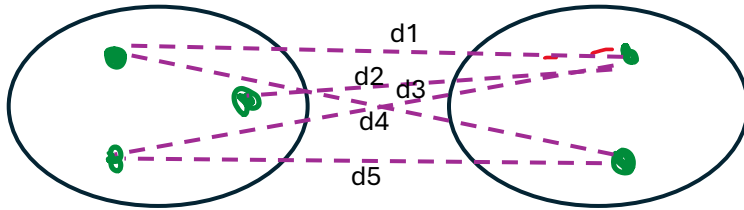
1. Single –link(Min)



1. Complete-link(Max)

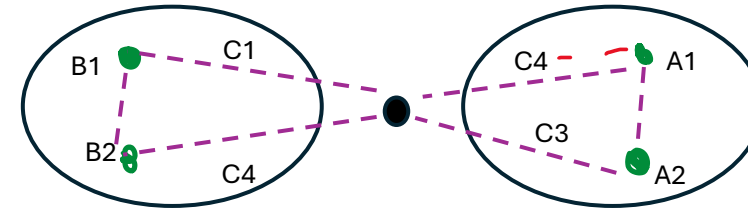


3. Group Average



$$\frac{d1 + d2 + d3 + d4 + \dots + dn}{\text{Total No. of points 1}^{\text{st}} \text{ \& 2}^{\text{nd}} \text{ cluster}}$$

4. Ward



$$\text{DIST} = (C1^2 + C2^2 + C3^2 + C4^2 + \dots + Cn^2) - (A1^2 + A2^2 + A3^2 + \dots + An^2)$$